

Hausübung 4

Aufgabe 1: $Q(P) = 4800 - 1200P$, $P = \frac{4800 - Q}{1200}$

$W = 400$, $C = 1.25 \cdot Q + 100$

a) $B = 900$ $Q = 800 + 400 = 1300$

$P = \frac{4800 - (q_1 + q_2)}{1200}$ $C = 1.25 \cdot (q_1 + q_2) + 100$

$P = 2.917$

$\pi_B = 1400.3$

$\pi_W = 566.8$

b) $MR = MC$

$R_B = P \cdot Q$ $MC = 1.25$

$= (4 - \frac{1}{1200} \cdot (q_1 + q_2)) \cdot q_1$

$= 4q_1 - \frac{1}{1200} \cdot (q_1^2 + q_1 q_2)$

$MR = 4 - \frac{1}{1200} \cdot (2q_1 + q_2) = 1.25$

$\frac{1}{1200} \cdot (2q_1 + q_2) = 2.75$

$2q_1 + q_2 = 3300$

$2q_1 = 3300 - q_2 \quad | :2$

$q_1 = \frac{3300 - q_2}{2}$

$q_1 = 1450 \checkmark$

c) $q_1 = \frac{3300 - q_2}{2}$

$q_2 = \frac{3300 - q_1}{2}$

$q_1 = \frac{3300 - \frac{3300 - q_1}{2}}{2}$

$q_1 = 1650 - \frac{1650 - \frac{q_1}{2}}{2}$

$q_1 = 1650 - 825 - \frac{q_1}{4}$

$q_1 = 825 + \frac{q_1}{4}$

$\frac{3}{4} q_1 = 825$

$q_1 = 1100$

$q_2 = 1100$

d) $\pi = P \cdot Q - C$

$= (4 - \frac{2200}{1200}) \cdot 1100 - 1.25 \cdot 1100 - 100$

$= 908.3 \checkmark$

Aufgabe 2: $Q(P) = 100 - \frac{1}{3}P$; $P = \frac{100 - Q}{1/3}$

$TC = 150 + 2q_i$

a) $\pi_i = \frac{100 - (q_1 + q_2)}{1/3} \cdot q_1 - 150 - 2q_1$

$\pi_1 = (300 - 3 \cdot (q_1 + q_2)) \cdot q_1 - 150 - 2 \cdot q_1$

$= 300q_1 - 3q_1^2 - 3q_1q_2 - 150 - 2 \cdot q_1$

$\pi' = 300 - 6q_1 - 3q_2 - 2 = 0$

$298 = 6q_1 + 3q_2$

$q_1 = \frac{298 - 3q_2}{6}$

$q_2 = \frac{298 - 3q_1}{6} \checkmark$

b) $q_1 = \frac{298 - 3 \cdot 20}{6} = 39.667 \checkmark$

c) $q_1 = \frac{298 - 3 \cdot \frac{298 - 3q_1}{6}}{6}$

$q_1 = q_2 \rightarrow q = \frac{298 - 3q}{6}$

$6q = 298 - 3q$

$9q = 298 \quad | :9$

$q = 33.11 \checkmark$

$Q = 66.22$

$\pi = 6428.29$

$\pi_1 = 3214.15$

$\pi_2 = 3214.15$